

1. Web-Based Experiment Platform

Backstory

Research in cognitive science and psychology has historically been confined to controlled laboratory settings, relying on specialized, expensive equipment and desktop software. This approach, while precise, severely limits the scale and demographic diversity of participant pools, often leading to findings that are not generalizable. Furthermore, the technical complexity of deploying millisecond-sensitive experiments on the web has been a significant barrier for researchers, preventing them from accessing larger, more representative samples online.

Problem Statement

Develop a secure, flexible, and high-precision Software-as-a-Service (SaaS) platform that empowers researchers to build and deploy complex behavioral experiments directly in a web browser. The platform's mission is to democratize cognitive science research by providing an accessible, no-code/low-code solution that replicates the accuracy of lab-based studies while enabling large-scale, diverse online data collection.

Key Features

- **High-Precision Timing Engine:** The core of the platform is a sophisticated engine meticulously engineered for millisecond-accurate stimulus presentation and response time measurement. This is critical for cognitive tasks and addresses the notorious difficulty of achieving reliable timing in a variable web environment.
- **Visual Experiment Builder:** A powerful no-code/low-code interface that allows researchers to visually design complex experimental trials. It supports features like custom logic, conditional branching, and advanced randomization of stimuli and trials, all without requiring the researcher to write any code.
- **Secure & Anonymous Data Handling:** A robust backend designed from the ground up to manage participant data. The system ensures participant anonymity through unique identifiers and is built to comply with strict research ethics standards (like IRB requirements and GDPR), guaranteeing data integrity and privacy.

2. AI-Powered Interview Bot

Backstory

Hiring has traditionally relied on human-led screening interviews that are time-consuming, costly, and highly inconsistent. The same answer can be judged very differently depending on the interviewer, which makes early-stage hiring decisions unreliable and often unfair. At the same time, candidates have very limited access to realistic interview practice and meaningful feedback. With remote hiring becoming the norm, ensuring the authenticity of an interview has become another major challenge, as there is little visibility into whether a candidate is responding on their own.

Problem Statement

Develop an intelligent AI-powered interview platform that can conduct personalized, human-like interviews at scale. The platform should understand a candidate's background, engage them in a natural spoken conversation, adapt its questioning based on their responses, and evaluate their performance in a fair and consistent manner. The mission is to make early-stage hiring faster, more objective, and more trustworthy for recruiters, while giving candidates an accessible way to practice and improve.

Key Features

- **Profile-Aware Questioning and Evaluation:** The platform understands a candidate's resume in the context of the target role and generates relevant, personalized interview questions. At the end of the session, it produces a comprehensive evaluation report that highlights the candidate's strengths, gaps, and overall suitability for the role.
- **Conversational and Adaptive Interviewing:** The interview is conducted as a natural spoken conversation rather than a static questionnaire. The platform intelligently adjusts the depth and direction of the interview based on each response, probing further where the candidate shows strength and adapting where they struggle, closely replicating the experience of a real interviewer.
- **Interview Integrity Monitoring:** The platform observes the candidate's attention throughout the session and identifies behaviour that may indicate the use of external help, such as repeatedly looking away from the screen. These observations are presented transparently to recruiters, building trust in the authenticity of every interview.